

★ WEEK 15 · PART 1 of 3

Bayesian State-Space: Why Point Estimates Lie

Honest Uncertainty for Dynamic Financial Models

Two layers of doubt · one joint posterior · a line versus a fan

We point the MCMC engine of Week 14 at the dynamic models of Weeks 8-10. A filtered point estimate hides its own uncertainty; a Bayesian state-space model gives a full posterior over hidden states and parameters alike.

1

PART 1 · WHY POINT ESTIMATES LIE

From a Confident Line to an Honest Fan

The quant meltdown that punished false confidence, the two layers of uncertainty in a dynamic model, the Bayesian state-space model that captures both, and the joint posterior over states and parameters.

The Quant Quake of August 2007 — Part 1

When confident point estimates met an uncertain market

In early August 2007, many quantitative market-neutral funds suffered sudden, severe, simultaneous losses. Their models — fit to confident point estimates — had not accounted for how crowded and fragile their shared positions really were.

What happened

- Quant long-short funds lost heavily over a few days.
- A rapid deleveraging cascaded through crowded trades.
- Then a partial, whipsaw rebound followed.

The uncomfortable truth

- The models looked precise and well-fit.
- They had no sense of their own fragility.
- Confidence, not the strategy, was the flaw.

The Quant Quake of August 2007 — Part 2

The lesson: model what you don't know

The episode became a lasting lesson: a model that reports one confident number, with no error bars, invites disaster. Honest quantitative work must quantify what it does NOT know — and let that uncertainty shape every decision.

The lesson

- A single estimate hides the range of what's possible.
- Unmodeled uncertainty doesn't vanish — it bites later.
- Quantify doubt; don't paper over it.

How this week responds

- Put full posteriors on dynamic models.
- Report credible bands, not lone lines.
- Size positions to real, stated uncertainty.

Bringing Two Threads Together

Where the whole course converges

1

We have dynamic models

Kalman and state-space models (Weeks 8-9) and HMMs (Week 10) track hidden states that evolve over time.

2

And we have Bayesian computation

Bayesian inference (Week 13) and MCMC (Week 14) give full posteriors instead of single point estimates.

3

This week joins them

Point the MCMC engine at the dynamic models, and get one joint posterior over hidden states AND parameters — with honest bands on everything.

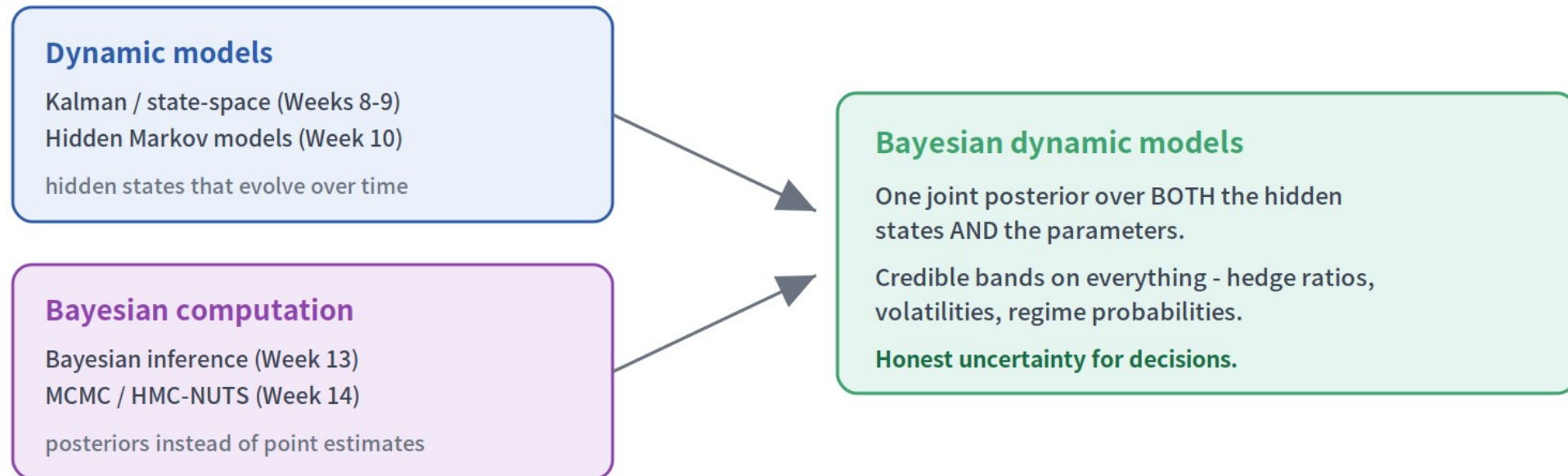
Part 1 in one line: a filtered line hides its uncertainty; a Bayesian state-space model reveals it.

Two Threads of the Course, Joined

The dynamic models of Weeks 8-10 supply hidden states; the Bayesian tools of Weeks 13-14 supply posteriors. Together they give Bayesian dynamic models — one joint posterior, with credible bands on every quantity.

Two Threads of the Course, Joined

Point the Bayesian engine at our dynamic models - and get honest uncertainty over hidden states.



The Gap in Point-Estimate Fitting

Why an MLE-filtered line can lie to you

Fitting a dynamic model by maximum likelihood and then filtering gives a single, smooth line for the hidden state. It looks authoritative — but it fixes the parameters at one estimate and ignores that they, too, are uncertain.

What the line does

- It tracks the hidden state through time.
- It looks clean, precise, and confident.
- It is a single best guess at each moment.

What the line hides

- The parameters were pinned at one estimate.
- Their uncertainty is nowhere in the picture.
- So the line looks far surer than it should.

Why a Filtered Line Looks Too Confident

Fixed parameters, false precision

The hidden assumption

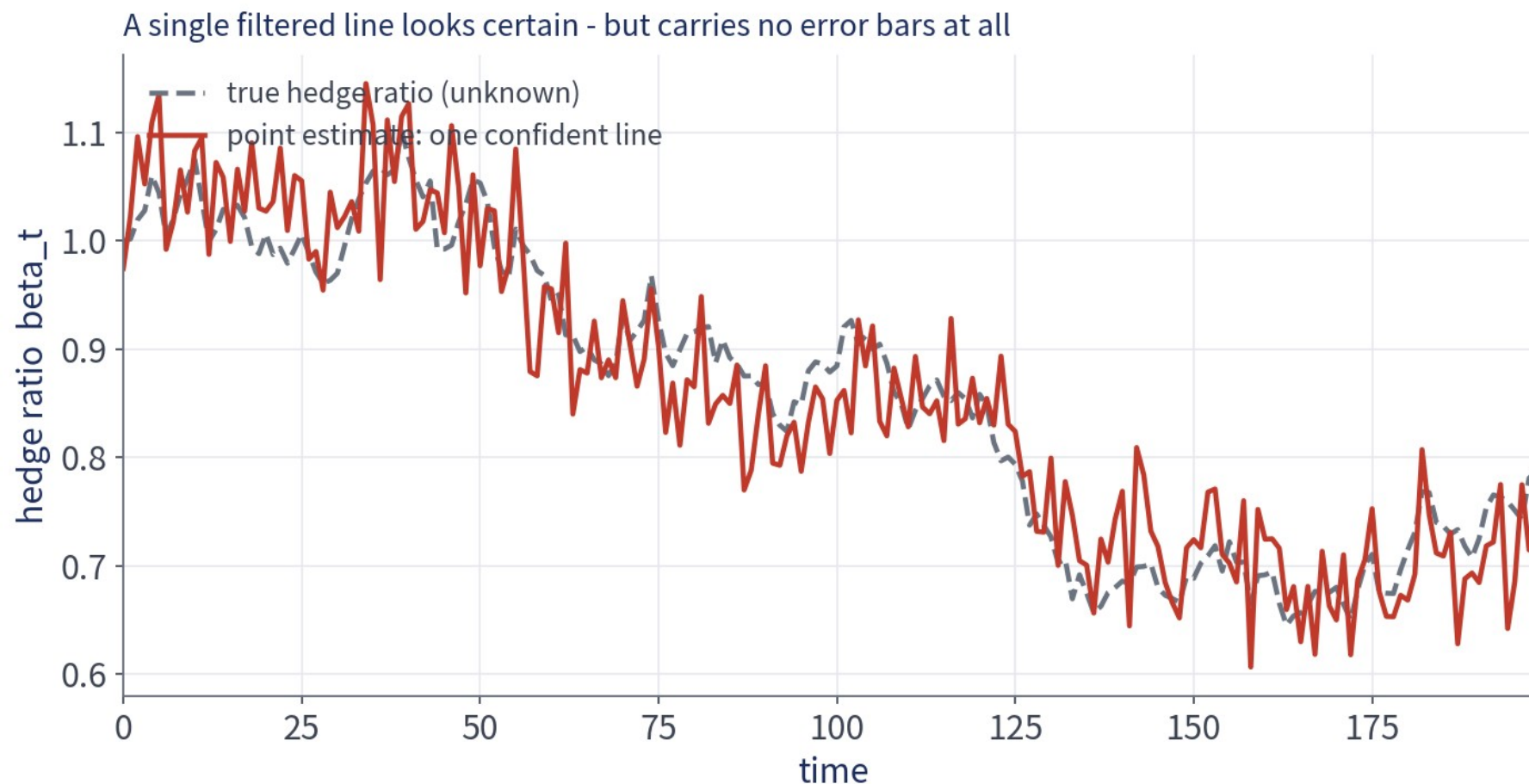
- The filter treats the estimated parameters as exact.
- It reports uncertainty about the state given those values.
- But those values were themselves just estimates.

The consequence

- The reported band is too narrow.
- Decisions built on it are overconfident.
- It is exactly the trap the quant quake exposed.

The Overconfident Point Estimate

A single filtered line for a time-varying hedge ratio looks authoritative and tracks the truth loosely — but it carries no error bars at all. Nothing on the chart tells you how much to distrust it.



Two Layers of Uncertainty

Where doubt comes from in a dynamic model

Honest uncertainty in a dynamic model has two layers. The first is state noise — the hidden state itself moves randomly. The second is parameter uncertainty — we don't know the model's parameters exactly. A truthful band must include both.

Layer 1 — state noise

- The hidden state evolves with random shocks.
- Even known parameters leave the state uncertain.
- This is what a filter already reports.

Layer 2 — parameter uncertainty

- The parameters are estimated, not known.
- Different plausible parameters give different states.
- A point-estimate filter ignores this layer entirely.

State (Process) Noise

The hidden state wanders on its own

What it is

- The state equation includes a random shock each step.
- So the hidden quantity drifts unpredictably.
- The hedge ratio, the volatility, the regime — all move.

How it shows up

- Even with perfect parameters, the state is uncertain.
- The filter's band captures exactly this.
- It is real, but it is only half the story.

Parameter Uncertainty

We don't know the model's dials

What it is

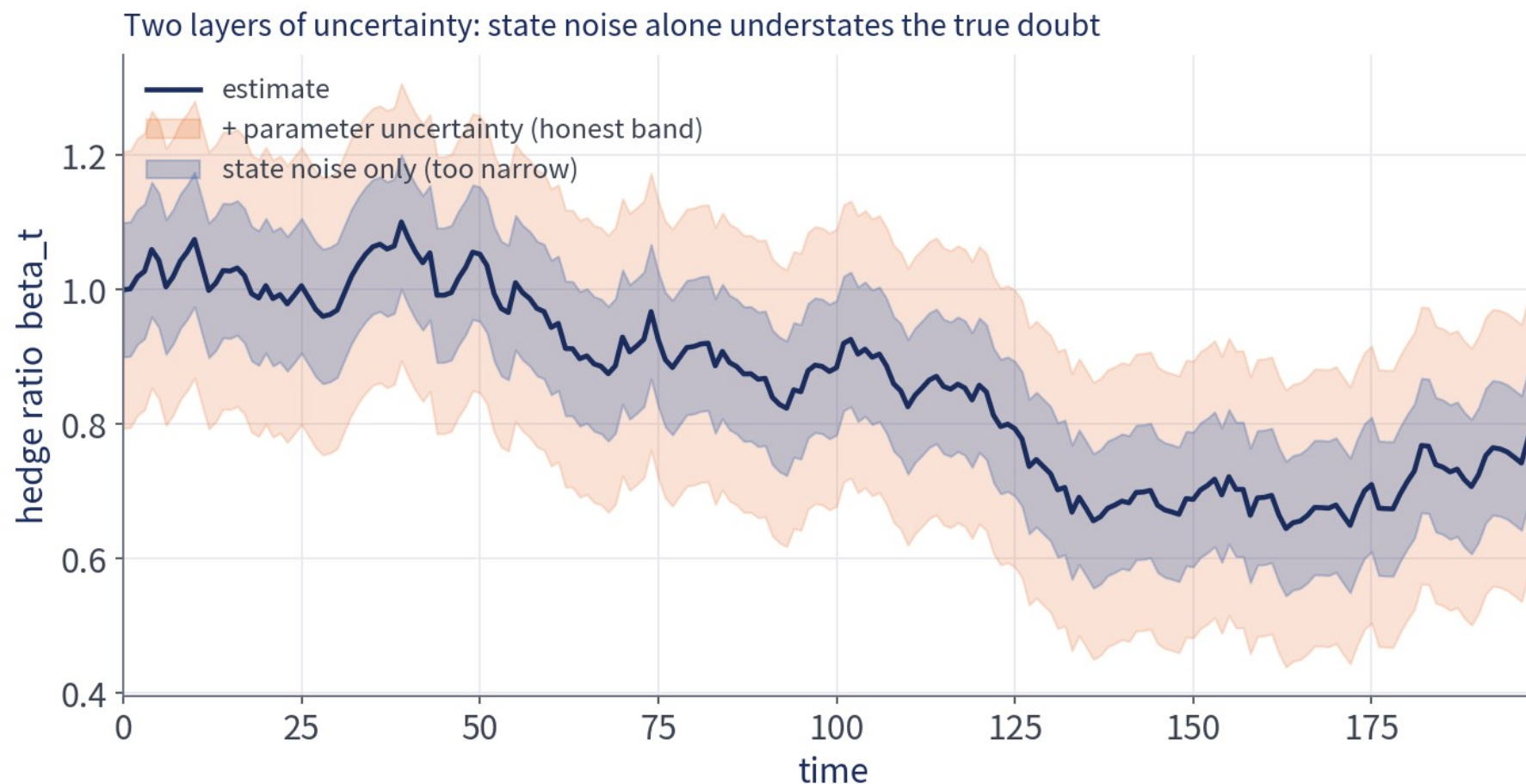
- The noise variances and coefficients are estimated.
- The data pin them down only approximately.
- Several parameter sets fit almost equally well.

Why it matters

- Each plausible parameter set implies a different state path.
- Ignoring this makes the band far too tight.
- The Bayesian approach carries it automatically.

The Two Layers, Side by Side

The inner band is state noise alone — what a filter reports. The wider band adds parameter uncertainty. The honest interval is the wide one; the narrow one quietly overstates how much we know.



The Honest Band Combines Both

One rule adds the two layers

$$\underbrace{\text{Var}(\beta_t \mid y)}_{\text{honest doubt}} = \underbrace{\mathbb{E}[\text{Var}(\beta_t \mid \theta, y)]}_{\text{state noise}} + \underbrace{\text{Var}(\mathbb{E}[\beta_t \mid \theta, y])}_{\text{parameter uncertainty}}$$

Reading it

- Total uncertainty splits into two pieces.
- The average state noise, plus the spread from parameters.
- This is the law of total variance.

The takeaway

- Drop the second term and the band is too narrow.
- A full posterior keeps both by construction.
- That is what makes the band honest.

The Bayesian State-Space Model

One joint posterior over states and parameters

The Bayesian state-space model treats the whole hidden state path AND the parameters as unknowns to infer together. We place priors on the parameters and let the data deliver a single joint posterior over everything at once.

What is unknown

- The entire path of the hidden state.
- The model's parameters — variances and coefficients.
- We infer them jointly, not one then the other.

What we add

- Priors that encode what we know beforehand.
- The same state and observation equations as before.
- The MCMC engine to explore the joint posterior.

The Model in Symbols

A time-varying coefficient, stated cleanly

$$\begin{aligned}\text{observation: } & y_t = \beta_t x_t + \varepsilon_t \\ \text{state: } & \beta_t = \beta_{t-1} + \eta_t\end{aligned}$$

The two equations

- Observation: the data equals a time-varying coefficient times a driver, plus noise.
- State: that coefficient drifts as a random walk.
- This is a dynamic linear model.

What is hidden

- The coefficient path is the hidden state.
- The noise variances are the parameters.
- Priors go on those variances.

The Joint Posterior

States and parameters, inferred together

$$p(\underbrace{\beta_{1:T}}_{\text{states}}, \underbrace{\theta}_{\text{params}} \mid y_{1:T}) \propto p(y_{1:T} \mid \beta_{1:T}, \theta) p(\beta_{1:T} \mid \theta) p(\theta)$$

Reading it

- The posterior is over the whole state path and the parameters.
- It factors into a likelihood, a state prior, and a parameter prior.
- MCMC samples from this joint distribution.

Why joint matters

- States and parameters inform each other.
- Inferring them together propagates both uncertainties.
- That is precisely the honest band we want.

What We Get From It

Credible bands on everything

Not a single answer

- A full posterior over the whole state path.
- A posterior over every parameter.
- Draws you can summarize any way you need.

Bands on real quantities

- A credible band around a time-varying hedge ratio.
- A band around a volatility or a regime probability.
- Uncertainty you can feed into decisions.

Point Estimate vs Full Posterior

One confident line, or an honest fan

Here is the whole shift in one image. The point estimate gives a single line for the hedge ratio. The Bayesian model gives a fan — a spread of plausible paths whose width is the honest uncertainty at every moment.

The line

- One value at each time.
- Clean, but silent about doubt.
- It hides how much it could be wrong.

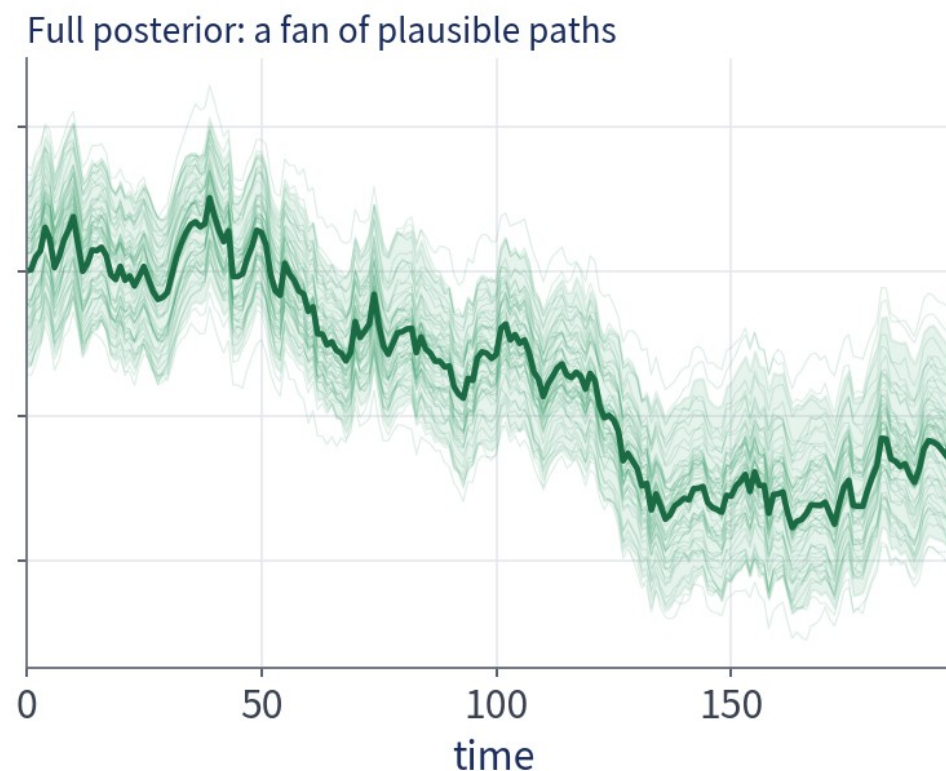
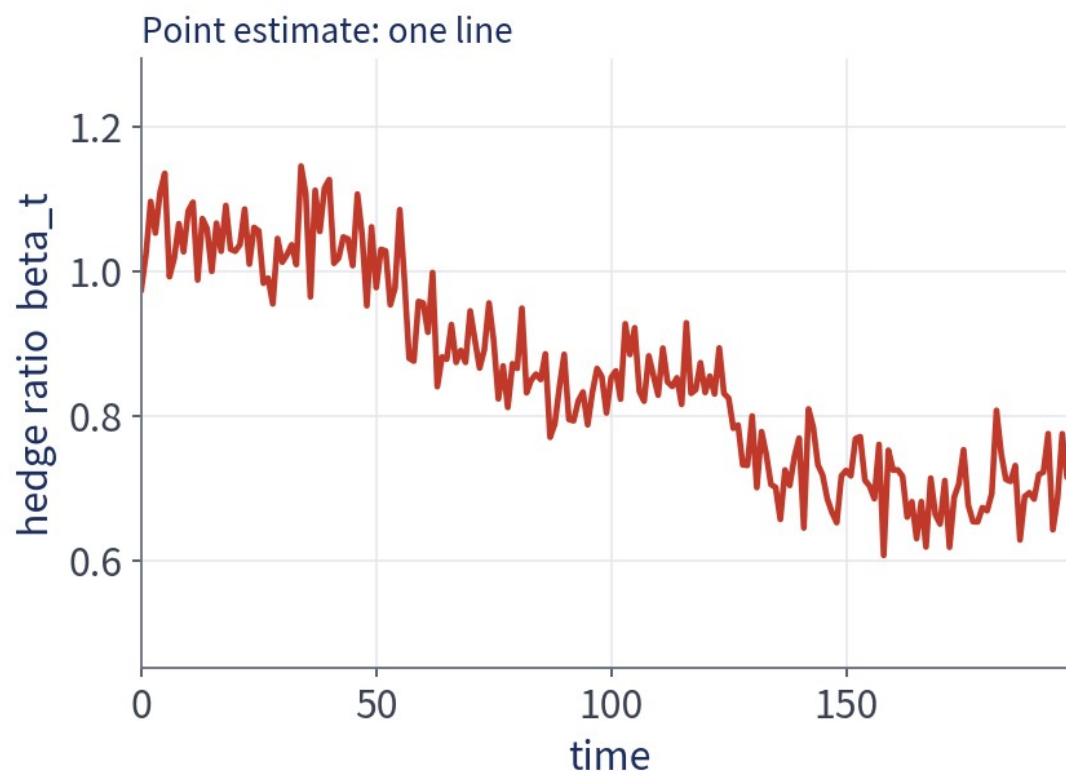
The fan

- A distribution of paths at each time.
- Its width is the uncertainty, stated plainly.
- Narrow where data is rich, wide where it is thin.

The Hedge Ratio: Line vs Fan

On the left, the point estimate is a single confident line. On the right, the posterior is a fan of plausible hedge-ratio paths with a credible band. The fan is the honest object to trade on.

The hedge ratio β_{t-1} : one confident line, or an honest fan of possibilities



Why the Fan Matters for Decisions

Uncertainty should shape the trade

With a line

- You act as if the estimate is exactly right.
- You size the same whether sure or unsure.
- You are blind to the risk of being wrong.

With a fan

- You see when the estimate is shaky.
- You can trade smaller when the band is wide.
- Uncertainty flows through to position size.

What Parts 2 and 3 Add

From the model to the machinery and the switches

Part 2 — sampling and applying

- Sampling states and parameters together.
- Forward-filter, backward-sample for whole paths.
- The Bayesian hedge ratio and stochastic volatility.

Part 3 — switching models

- The Bayesian HMM and soft regime probabilities.
- Structural breaks at uncertain dates.
- Trading with uncertainty — pairs Stage VI.

See the Overconfidence for Yourself

Direct Claude Code to compare a line and a fan

GOAL Fit a time-varying coefficient two ways and see how a point estimate understates the uncertainty.

[INPUT] Prompt · Claude Code

Write and run `line_vs_fan.py` that:

1. simulates a dynamic linear model - a hedge ratio that drifts as a random walk, with noisy observations;
2. fits it with a Kalman filter at point-estimate parameters and plots the single filtered line;
3. also draws many plausible paths accounting for parameter uncertainty, forming a fan with a band;
4. overlays both against the true hidden path.

Show the point band is too narrow and the fan covers the truth honestly.

How to work with it

- The line fixes the parameters.
- The fan carries their uncertainty.
- The fan is wider - and honest.
- It is narrow where data is rich.

Deliverable: `line_vs_fan.py` + the filtered line and the posterior fan against the true path.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1

What two layers make up honest uncertainty in a dynamic model?

**State noise +
parameter
uncertainty**

Q2

What does a Bayesian state-space model infer?

**A joint posterior over
states and
parameters**

Q3

What replaces the single confident line?

**A posterior fan with a
credible band**

All three come from the two layers, the joint posterior, and the line-versus-fan contrast.

Three Misconceptions to Avoid

Where state-space intuition slips

Common myths

- "A filter's band already shows all the uncertainty."
- "Point estimates are fine if the model fits well."
- "States and parameters can be fit one then the other."

The reality

- It shows state noise but omits parameter uncertainty.
- A good fit can still be dangerously overconfident.
- Inferring them jointly is what propagates both doubts.

Three Things to Carry Forward

Part 1 in one screen

1

Point estimates hide their uncertainty

A filtered line fixes the parameters at one estimate and reports only state noise. It looks precise but omits parameter uncertainty — the false confidence the quant quake punished.

2

Honest doubt has two layers

State noise and parameter uncertainty together make the true band. The law of total variance adds them; a full posterior keeps both by construction.

3

The Bayesian state-space model gives a fan

It infers one joint posterior over the hidden state path and the parameters, turning a single confident line into an honest fan with credible bands on everything.

Next — Part 2: sampling states and parameters together with FFBS, and the Bayesian hedge ratio.